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ACUTE PAIN

Erector Spinae Plane Block Recipe (For Rib Fractures)

YG CG 1105 Aug 2023
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Key Points

- This guideline supplements YG CG 1103: Management of Patients with Rib Fractures
- ESPB is considered first-line regional technique for these patients

Introduction

- Patients admitted with Rib fractures should be managed according to YG CG 1103 and our MTCs 'START' Pathway.
- Effective Opioid-sparing Multi-modal analgesia is pivotal in reducing risk of developing respiratory complications and optimising respiratory function.
- The Erector Spinae Plane Block is well suited to this approach and is currently considered first-line. The MTP (Mid-point of Transverse process to Pleura) block is a slightly deeper and thus, more advanced version.
- Alternative continuous techniques:
  - Serratous Anterior Block – More 'ergonomic' catheter entry point, Better suited than ESPB for antero-lateral rib fractures.
  - Paravertebral Block – More effective pain relief, but greater risk of pneumothorax. Well suited to patient with chest drains in situ.
  - Epidural Block- More effective pain relief, BUT, more invasive & mandates critical care admission.
- Additional blocks may be of benefit in the presence of scapula & acromio-clavicular fractures.

Pre-Procedure Assessment

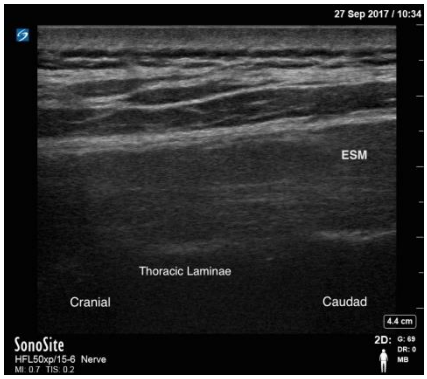
- Assess and document indication for procedure: chest injuries, medications, coagulation status, allergies & current pain severity including when coughing and breathing.
- There are no absolute contraindications except invalid consent.
- The risk of clinically significant haemorrhage or haematoma is very low, Thus, coagulopathy or the use of anticoagulants or antiplatelet drugs are not absolute contraindications to ESP block. However, an individualized risk-benefit assessment should be performed for every patient.
- Document assessment, indication for procedure and consent (Consent form & checklist required for patients attending theatre)
- Mark clearly the side of procedure.

Top Tips

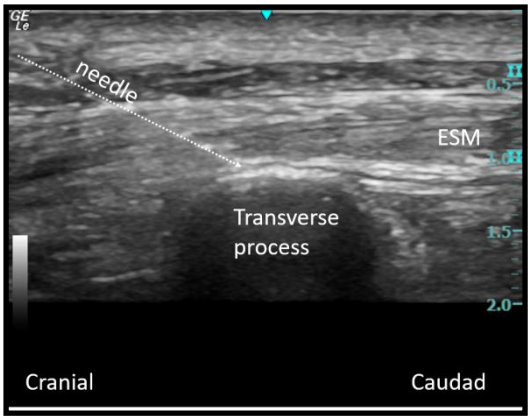
- Perform the procedure in AR 4 or in an ED resus/majors bay.
- Monitoring and Asepsis standards as per neuraxial procedures.
- Think Ergonomics! Position patient & kit comfortably for both of you.
- Choose the transverse process in the middle of the affected ribs.
- Inject 1-3 mL of local anaesthetic to confirm proper injection plane by seeing the erector spinae muscles 'lift off' the transverse process.
- Complete the nerve block with 30 mL of local anaesthetic.



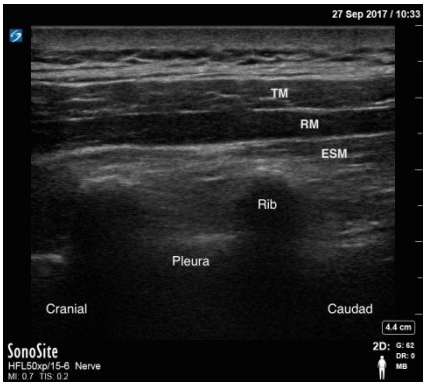
Visualisation:



Too Medial!



Perfect!



Too Lateral!

The MTP Block is a modification of this block, whereby LA is deposited at the mid-point between the posterior border of the transverse process and the pleura, followed by insertion of the catheter through the needle:

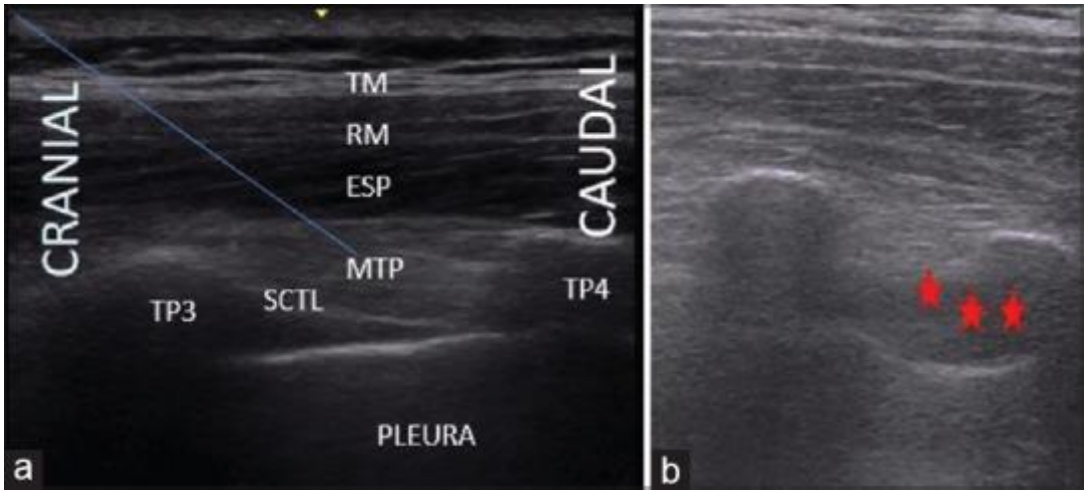


Image of US guided MTP block: Sagittal view - T3-T4 transverse process, (a) before local anesthetic (LA) deposition; arrow indicates needle direction from cranial to caudal. (b) after LA deposition ***LA spread.

LA Maintenance Regimen

If Surgical Ward: Elastomeric Pump + 1x Manual bolus

- 0.125% L-Bupivicaine 8ml per hour via Elastomeric Pump (Prescribe on front of drug chart)
- 10ml 0.25% L-Bupivicaine bolus 1-2x per day* (Prescribe in regular section of drug chart)

* Administered by 063 bleep holder.

If Critical Care: CADD-Solis 'Yellow Epidural' Pump

- 0.125% L-Bupivicaine 500ml. Set bolus: 15ml every 2 hours. Plus clinician controlled bolus: 10ml (Instruction below)
- N.B. the first programmed bolus, will be delayed by 2 hours to account for the bolus given on catheter insertion.

Setting up the CADD-Solis Pump Intermittent Bolus (PIB) & Clinician-Controlled Bolus (CCB)

- Switch pump on (Grey button on right).
- Screen displays “Do you want to start a new patient?” Press “Yes”.
- The “Select therapy” menu is displayed.
- Scroll down to “Regional Analgesia” Program’, press ‘select’.
- Select “Rib Fracture (ESPB/SAPB)”.
- The “Select Drug” menu is displayed. Confirm “0.125% Levobupivacaine”.
- Enter security code “007”.
- Check program. Press “Yes”.
- Review pump settings : Intermittent bolus (PIB) :Default is 15 ml. (Press ‘select’ to edit to 5 or 10mls, if appropriate.)
  Bolus Interval: 2 hours
  Reservoir volume: 500ml
- Press Accept Value.
- If cassette is not already attached, there will be a prompt to ‘Attach Cassette’.
- To ‘Prime tubing’, press ‘Prime’.
- Screen displays “Start pump?” Press “Yes”.

Administering a Clinician-Controlled Bolus

- Press ‘Task’ button.
- Select ‘Clinician Bolus’
- Enter ‘007’ security code.
- Press ‘Deliver’.

After-Care

The ESP catheter is kept in place as long as it is providing analgesic benefit. Remove if the site becomes infected, local anaesthetic leaks, before discharge or at the patient's request.

Patients should be assessed daily, focusing on ambulatory and breathing pain scores, 24hr opioid requirements, incentive spirometry and ambulation status.

Any troubleshooting is undertaken by Acute Pain staff and on call Anaesthetic team.

References

1. **American Society of Regional Anaesthesia:** How I Do It: Erector Spinae Block for Rib Fractures: The Penn State Health Experience (asra.com)
2. Erector Spinae Plane Nerve Block - NYSORA
3. Saudi J Anaesth. 2019 Oct-Dec; 13(4): 365–367.